Lab: Description Logics

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1. Basics

1) Why do we say Description Logics (DL) is a concept language? What does "description" mean in DL?

Description Logics are called concept languages because their primary purpose is to describe and structure concepts and the relationships among them in a domain, rather than to express arbitrary first-order logic statements. In description logics, descriptions denote collections of entities. Used for Identify definitions for terms, positions, and people involved. Descriptions are constructed by a set of logical constructors. Common constructors are: Concept name, role name, Boolean operators and quantifiers

2) In DL, what do the TBox and the ABox stand for?

In Description Logics (DLs), the TBox stands for terminology box, TBox contains statements about the concepts and roles in the domain. It defines the relationships between concepts and roles, and it specifies which concepts are subclasses of other concepts. The ABox stands for assertion box, The ABox contains statements about the individuals in the domain. It asserts that individuals belong to certain concepts and that they have certain roles with other individual.

3) AL is the base DL in the family of DLs. What are the constructors provided by this language? For each of the constructors, give its semantics using interpretation.

In AL, we use different constructors. Their meanings are:

Top ($\top^I$): This includes everything in the domain. Under interpretation I, it means the whole domain is included. ($\top^I = \Delta^I$)

Bottom ($\bot^I$): This means nothing, an empty group. Under interpretation I, it contains no elements. ($\bot^I = \emptyset$)

Atomic concept (A): interpreted as a subset of the domain ($A^I \subseteq \Delta^I$).

Negation of atomic concepts (¬A): represents the complement of A in the domain ($\Delta^I \setminus A^I$).

Conjunction (C ⊓ D): This shows objects that belong to both C and D at the same time. Under interpretation I, it is the overlap of C and D. ($C^I \cap D^I$)

Universal restriction (∀R.C): This means all related items through role R must be in concept cruder interpretation I, every R connection from x must go to something in C.

Limited existential restriction (∃R.A): This means there is at least one connection through R to something in class A (only atomic concepts allowed).Under interpretation I, x must have at least one R link to an element in A.

4) We obtain more expressive description languages if we add further constructors to AL. List at least 3 constructors. For each constructor, give its interpretation and the letter that indicates the constructor.

The letter U, union of concept: it can be interpreted as

$(C \sqcup D)I = CI \sqcup DI$

The letter £, full existential quantification: it can be interpreted as

$(\exists R.\ T)I = \{a \in \Delta I \mid \exists b.\ (a, b) \in RI\}$.

The letter N, for number restrictions: it can be interpreted as

$(\leq n\ R)\mathbf{I} = \{ a \in \Delta \boldsymbol{I} \mid \mid \{b \mid (a,b) \in \boldsymbol{RI}\} \mid \geq n\}$

$(\leq n\ R)I = \{ a \in \Delta I \mid\mid \{b \mid (a,b) \in RI\} \mid \leq n\}$

5) Inclusion and equality are the two forms of terminological axioms in DL. What are the differences between these two axioms?

Inclusion (C ⊑ D): This means every C is also a D, but not necessarily the other way around.

Equality (C ≡ D): This means C and D are exactly the same, and share all members

6) In DL, what are the four decidable properties of TBox? What are the reasoning tasks for ABox?

Four decidable properties of Tbox are satisfiability, subsumption , equivalence and disjointness.Some common reasoning tasks for Abox includes consistency checking, instance checking.

2. Translate the DL statements into Natural Language

1) Person ⊓ Happy
A person who is happy.

2) Person ⊓ Happy ⊓ ∃owns.Pet
A happy person who owns at least one pet.

3) Person ⊓ ∀owns.Cat
A person whose pets are only cats.

4) Person ⊓ ¬Happy ⊓ ∃owns.(Cat ⊓ Old)
A person who is not happy and owns at least one old cat.

5) Person ⊓ ∃owns.Pet ⊓ ∀owns.(Cat ⊔ Fish)
A person who owns at least one pet, and all of their pets are either cats or fish.

3. DL Formalization

1) Women are persons.
Woman ⊑ Person

2) Men are persons.
Man ⊑ Person

3) Johan is a man.
Man(Johan)

4) Bill is a man.
Man(Bill)

5) James is a man.
Man(James)

6) Johan is not Bill.
Johan ≠ Bill

7) Johan is James.
Johan ≈ James

8) A parent is a person and has a child.
Parent ≡ Person ⊓ ∃hasChild.Person

9) Bill is a child of Johan.
hasChild(Johan, Bill)

10) A sibling is either a brother or a sister.
Sibling ≡ Brother ⊔ Sister

11) Alice is her own sibling.
hasSibling(Alice, Alice)

12) A person is their own relative.
Person ⊑ ∃isRelativeOf.Self

13) Every grandfather is a man and has a child who is a parent.
Grandfather ≡ Man ⊓ ∃hasChild.Parent

14) A childless person is a person but not a parent.
ChildlessPerson ≡ Person ⊓ ¬Parent

15) An unmarried person is a person who does not have a spouse.
Unmarried ≡ Person ⊓ ¬∃hasSpouse.Person

16) A happy person has only happy children.
Happy ⊑ Person ⊓ ∀hasChild.Happy

17) A great-grandfather is a father with more than 4 children and all children are parents.
GreatGrandfather ⊑ Man ⊓ (≥5 hasChild.Person) ⊓ ∀hasChild.Parent

18) A child is the inverse of a parent.
Child ≡ ∃hasChild⁻.Person

19) Being a father is a sub-role of being a parent.
hasFather ⊑ hasChild

20) A grandparent is a parent's parent.
Grandparent ≡ ∃hasChild.Parent

21) A child cannot be both son and daughter at the same time.
Son ⊓ Daughter ⊑ ⊥

22) A parent has a child if and only if they have a son or a daughter.
∃hasChild.Person ≡ ∃hasChild.Son ⊔ ∃hasChild.Daughter

4. Create a Knowledge Base for University using SROIQ

<u>Primitive Concepts:</u>

Concept: University

Concept: Department

Concept: Course

Concept: Student

Concept: Professor

Concept: ResearchProject

Concept: Assignment

Concept: Exam

<u>Defined Concepts:</u>

Concept: GraduateStudent

Concept: ResearchStudent

Concept: HeadOfDepartment

Concept: Supervisor

<u>Roles:</u>

Role: teaches

Role: enrolledIn

Role: offers

Role: worksIn

Role: supervises

Role: manages

Role: hasAssignment

Role: hasExam

Role: hasProject

Role: hasAdvisor

TBox Axioms:

University ≡ ∃offers.Course

Department ≡ ∃offers.Course ⊓ ∃partOf.University

Professor ≡ ∃teaches.Course ⊓ ∃worksIn.Department ⊓ (≥1teaches.Course)

Student ≡ ∃enrolledIn.Course

GraduateStudent ≡ Student ⊓ ∃enrolledIn.GraduateProgram

ResearchStudent ≡ Student ⊓ ∃hasProject.ResearchProject ⊓ ∀ hasAdvisor.Professor

HeadOfDepartment ≡ Professor ⊓ ∃manages.Department

Supervisor ≡ Professor ⊓ ∃supervises.ResearchProject

ABOX

University(JU).

Department(DeptAI).

Course(TAAI).

Student(Mincy).

Professor(DrRobert).

ResearchProject(Project1).

Assignment(A1).

Exam(E1).

Role Assertions:

offers(JU, TAAI).

worksIn(DrRobert, DeptAI).

teaches(DrRobert, TAAI).

enrolledIn(Mincy, TAAI).

hasAssignment(TAAI, A1).

hasExam(TAAI, E1).

hasProject(Mincy, Project1).

supervises(DrRobert, Project1).